

Physics Packet: Static & Kinetic Forces Review

Name: _____

Date: _____

Part A: Forces on a Box at Rest

A box rests on a level surface and is **not moving**.



1. Draw the Free Body Diagram (FBD)
2. What are my x forces?
3. What are my y forces?
4. What is $\sum F_x$?
5. What is $\sum F_y$?

Part B: Static Friction

A student **pushes the box to the right**, but the box does **not move**.



1. Draw the Free Body Diagram (FBD)
2. What are my x forces?
3. What are my y forces?
4. What is $\sum F_x$?
5. What is $\sum F_y$?

3) Since the box is not moving, what must be true?

Part C: Maximum Static Friction

As the student pushes harder, static friction increases to resist motion.



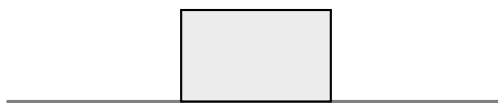
1. Draw the Free Body Diagram (FBD)
2. What are my x forces?
3. What are my y forces?

4. What is $\sum F_x$?

5. What is $\sum F_y$?

Part D: Kinetic Friction

A force is applied to the box causing it to move.



1. Draw the Free Body Diagram (FBD)

2. What are my x forces?

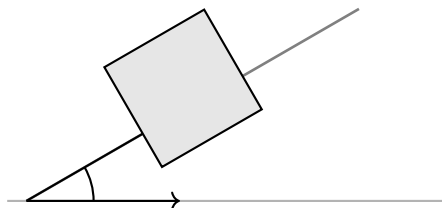
3. What are my y forces?

4. What is $\sum F_x$?

5. What is $\sum F_y$?

Part E: Forces on an Incline Plane

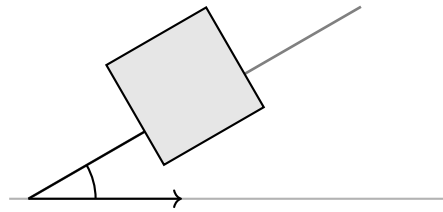
A box is on an flat incline making an angle θ with the horizontal.



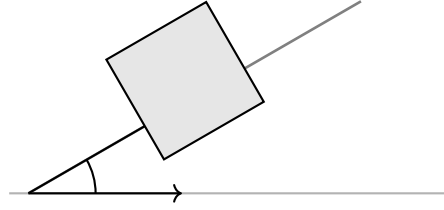
1. Draw the Free Body Diagram (FBD)
2. What are my x forces?
3. What are my y forces?
4. What is $\sum F_x$?
5. What is $\sum F_y$?

Part F: Forces on an Incline Plane Static Friction

A box is on an flat incline making an angle θ with the horizontal.



1. Draw the Free Body Diagram (FBD)
2. What are my x forces?
3. What are my y forces?
4. What is $\sum F_x$?
5. What is $\sum F_y$?



Part G: Forces on an Incline Plane Kinetic Friction

A box is on an flat incline making an angle θ with the horizontal.

1. Draw the Free Body Diagram (FBD)
2. What are my y forces?
3. What are my x forces?
4. What is $\sum F_x$?
5. What is $\sum F_y$?

Summary for all friction:

1. What is the formula for static friction?
2. What is the formula for kinetic friction?
3. What does $F_p = F_{f_s} = \mu_s F_N$ mean?
4. What does $F_p = F_{f_s} < \mu_s F_N$ mean?

5. What does $F_p > \mu_s F_N$ mean? What formula do we use now?
6. On a flat surface, when there is **no push force**, and a box **is at rest** what are $\sum F_x$ and $\sum F_y$?
7. On a flat surface, when there is a **push force**, and a box **is at rest** what are $\sum F_x$ and $\sum F_y$?
8. On a flat surface, when there is a **push force**, and a box **is moving** what are $\sum F_x$ and $\sum F_y$?
9. On an inclined plane, when there is **no push force**, and a box **is at rest** what are $\sum F_x$ and $\sum F_y$?
10. On an inclined plane, when there is a **push force**, and a box **is at rest** what are $\sum F_x$ and $\sum F_y$?
11. On an inclined plane, when there is a **push force**, and a box **is moving** what are $\sum F_x$ and $\sum F_y$?